



# MANIPAL

ACADEMY of HIGHER EDUCATION

*(Institution of Eminence Deemed to be University)*

**MANIPAL SCHOOL OF INFORMATION SCIENCES**

**(A Constituent unit of MAHE, Manipal)**

## **AI - Based Intrusion Detection System for IoT Networks Using Machine Learning and Deep Learning**

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## **ABBREVIATIONS**

|     |                             |
|-----|-----------------------------|
| ADC | Analog to Digital Converter |
| ADK | Arduino Development Kit     |
| AI  | Artificial Intelligence     |

## 1. Introduction

IoT devices became one of the most core used devices and which have helped in the advancement to smart homes but at the same time there are some issues from that which can be like the data being too open or being vulnerable and due to that there are chances that the data might be leaked.

The core issue of these IoT devices is the security which is very weak and can be easily penetrated and due to this the attacker or the intruder can easily get a hold on some of the sensitive information that currently happens at the home and he can even intercept the connection and can tamper the data that flows from one point to other which might lead to major consequences.

The layers in IoT can be either three, four or five but the scientists have decided to keep it down to three layers and due to that we have perception layer, network layer and application layer and these are the main places where the attacks are tend to happen and they can be of different types such as in case of perception layer the actuators data can be tampered or can be jammed using various methodologies and in case of network layer the data can be heard by either man in the middle attack or even the attacker can do a spoofing and can just keep listening to the whole conversation and the attacker will even have the network mapping of that particular place or the device which gives him an idea of what all are the devices that are connected and how can it be interrupted and then at last comes the application layer and the data in this can be attacked with the help of DDOS, asynchronization and so and due to this the message that has been sent on a side and had to be received on other side might face a delay and due to asynchronization it can lead to either weak internet connection or all the messages might get transmitted at once.

The main issue has been that there are already some methods to get hold of the intruders and some of them being the signature based model where the model detects which is an attacker and which is an intruder based on the previous trends but this can lead to an issue because lets consume a scenario where the data that has been given has around 90% of anomaly and 10% of normal data and if in case the entry in the network flow is a normal one, even then that particular one will be marked as anomaly because of the earlier trend and it will remember that all the entries or the entire traffic is made up of anomaly so to keep it simple, in case of signature based detection, the detection will just keep everything under the category which has majority value and in the above case it can be seen that anomaly is 90% while the normal data is only

around 10% and due to this even though if the data that enters the network is a normal safe user, even that will be marked as anomaly.

The next one is a rule based detection and in this case, it becomes hard because these rules are written by the scientist or the user and each and every time, a new attack is found then the rules have to be changed and due to this it becomes really hard for the user or the scientist to work on it and keep the rules changing for each and every new entry.

There are many aspects which have to be considered if a detection system is to be built and for that